

KTQ.

$$1. KE = \frac{3}{2} nRT = \frac{1}{2} KT = \frac{1}{2} fKT = \frac{1}{2} f(KT)(N_A)$$

2. Internal Energy (U)

$$KE \text{ of } n \text{ molecules GAS} = \frac{n}{2} fRT$$

where

$n \rightarrow$ No. of moles

$R \rightarrow$ Constant mono, di, Tri Atomic mix

$T \rightarrow$ Temp

$f \rightarrow$ degrees of freedom

$K \rightarrow$ Boltzmann's constant.

$N_A \rightarrow$ Avagadro's No.

$$Rv = \frac{1}{3} mc^2$$

Mono $3T + 0R$

Di $3T + 2R$

Poly $3T + 3R$

$$C = \frac{fR}{2}$$

"Graham's Law"

$$\sigma \propto \frac{1}{\sqrt{d}}$$

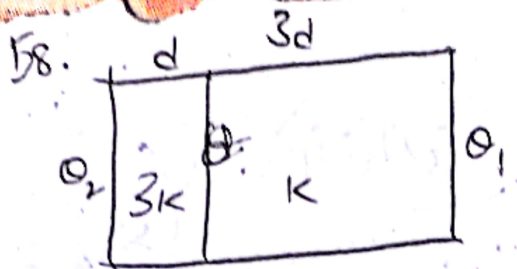
$\sigma \rightarrow$ rate of Effusion

$d \rightarrow$ density...

$$3. \quad R_{ms} \quad v = \sqrt{\frac{3KT}{M}}$$

$$Avg \quad v = \sqrt{\frac{8KT}{\pi M}}$$

$$Most \text{ Probable } v = \sqrt{\frac{2KT}{M}}$$



$$C = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3KT}{m}}$$

Where.

$m \rightarrow$ Molecular Mass

$M \rightarrow$ Mass of Molecule

$$Z = 2 + 2 = 4$$

$$9Q_2 - 9Q_1 = 9Q_2 - 9Q_1$$

$$9Q_2 - 9Q_1 = 9Q_2 - 9Q_1$$

$$\frac{1}{K_A} = \frac{1}{K_A} + \frac{1}{K_A}$$

Translatory KE of Gas Molecules

$$= \frac{1}{2} m c^2$$

$$\frac{4d}{K(A)} = \frac{d}{3KA} + \frac{3d}{KA} = \frac{4d}{KA} = \frac{1}{K} (3 + 1) = \frac{4}{K}$$

$$\frac{4}{K} = \frac{1}{3K} + \frac{3}{K} = \frac{1}{K} (1 + 3)$$

Gases:

Monoatomic $3T = 3$

Diatomic $3T + 2R = 5$

Polyatomic $3T + 3R = 6$

$$\frac{2}{K} = \frac{1}{K} (1 + 1)$$

$$\frac{6K}{5} = R \epsilon_1 \quad \frac{d\epsilon}{dt} = \frac{T_2 - T_1}{T_2} = \frac{5}{6} (1 - 0)$$

$$C = \frac{fR}{2}$$

$$\frac{d\epsilon}{dt} = \frac{-K \epsilon (T_1 - T_2)}{1}$$

C_v	(C)	$\gamma = \frac{C_p}{C_v}$
$\frac{3R}{2}$	$\frac{5R}{2}$	$\frac{5}{3}$
$\frac{5R}{2}$	$\frac{7R}{2}$	$\frac{7}{5}$
$3R$	$4R$	$\frac{4}{3}$

i. KTG.

ii. Thermodynamics

"KTG"

i) Microscopic

ii) Macroscopic

RMS KE of Gas Particles

Pressure Volume Moles

$$P = \frac{1}{3} m c^2$$

Where.

$m \rightarrow$ Total Mass of Gas

$C \rightarrow$ RMS Velocity

$P \rightarrow$ Pressure

$V \rightarrow$ Volume

Law of Equipartition:

i) KE of Gas Molecule = $\frac{1}{2} K T$

ii) KE of Gas Molecule in f Degrees of freedom = $\frac{f}{2} K T$

iii) KE of one mole of Gas Molecule = $\left(\frac{f}{2} K T\right) N_A$

Where.

$K \rightarrow$ Boltzmann's Constant

$N_A \rightarrow$ Avogadro's No.

$$RMS V_{RMS} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3KT}{m}}$$

Translatory Velocity Only...

iv) KE of n Mole of Gas = $\frac{n}{2} f R T$

Internal Energy (U)

$\sqrt{\frac{3KT}{m}}$
 → Molecular Mass
 → Mass of Molecule
 ory KE of Gas Molecules
 $\frac{1}{2} m c^2$

$\left(\frac{3KT}{m} \right) = \frac{3KT}{2}$

omic $3T = 3$ Temp constant
 mic $3T + 2R = 5$
 omic $3T + 3R = 6$

$\left(\frac{R}{2} \right) \times \times \times$

$\gamma = \frac{C_v}{C}$
$\frac{5}{3}$
$\frac{7}{5}$
$\frac{4}{3}$

fition :
 Molecule. $= \frac{1}{2} KT$
 Molecule. egress of $= \frac{f}{2} KT$
 2dom
 or of $= \left(\frac{f}{2} KT \right) (N_A)$
 icule

Mann's Constant
 ad's NO.

ob Gas $= \frac{n}{2} fRT$

ergy (u)

$K N_A = R$
 $U = \frac{n}{2} fRT$
 $U = nC_v T$
 $\Delta U = nC_v \Delta T$

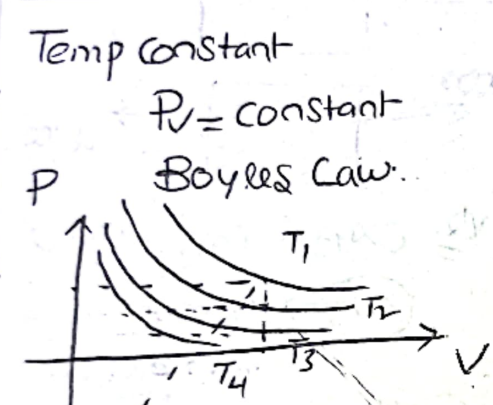


Internal Energy $\propto T$

$T \uparrow U \uparrow$
 $T \downarrow U \downarrow$
 $T = \text{constant } U = 0$

Change in internal Energy ...

GAS LAWS:



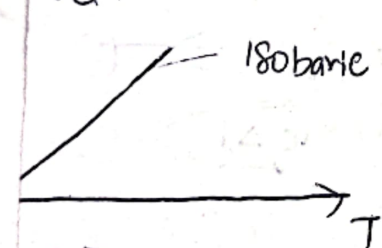
$T_1 > T_2 > T_3 > T_4 > T_5 \dots$ isothermal lines...

Hint 1:

Draw imaginary Lines... or
 Do by $PV = nRT$

$PV = nRT$

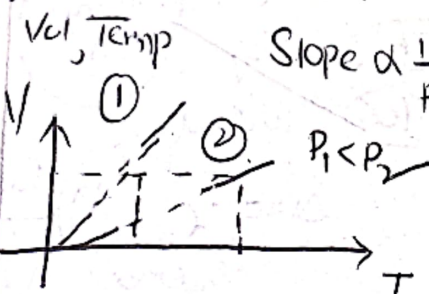
Volume constant
 $V \propto T$



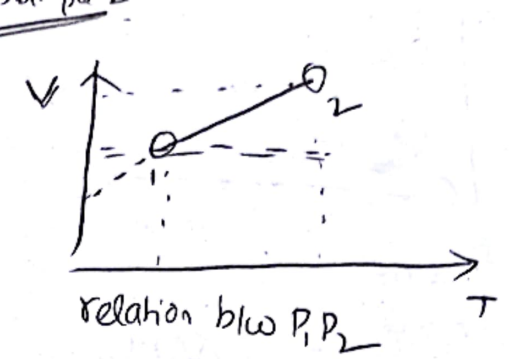
$P_1 = P_2$

Vol, Temp

HINT 2:

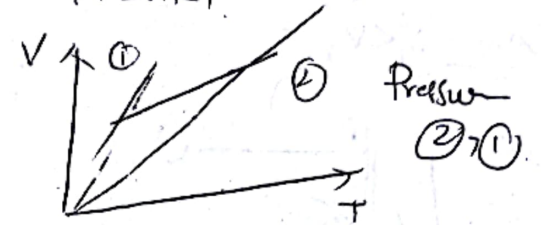


Slope $\propto \frac{1}{P}$
 $P_1 < P_2$



relation b/w P_1, P_2

$PV = nRT$

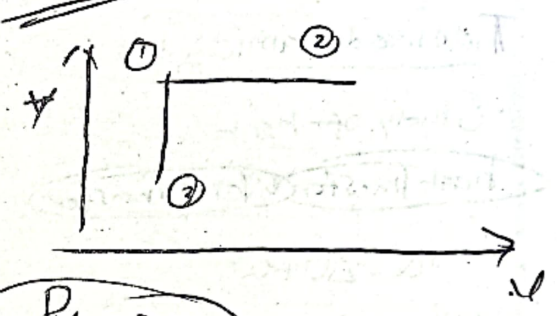


Pressure
 $(2) > (1)$

Hint 1, 3

Discontinuous Graph then draw
 Imaginary Lines From origin

Example 3



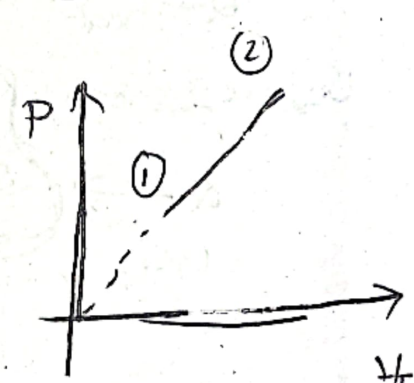
$PV = nRT$

$P_1 > P_2$
 $P_1 > P_3$

3. $V = C$

$P \propto T$

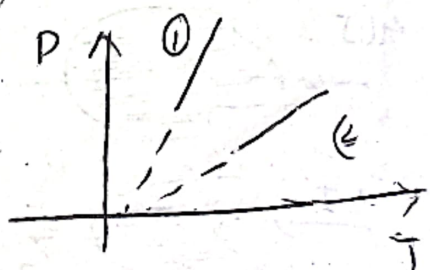
$V_1 = V_2$



$PV = nRT$

$V = \frac{nRT}{P}$

$V \propto \frac{1}{\text{Slope}}$



$V_2 > V_1$
 $\frac{1}{2}$